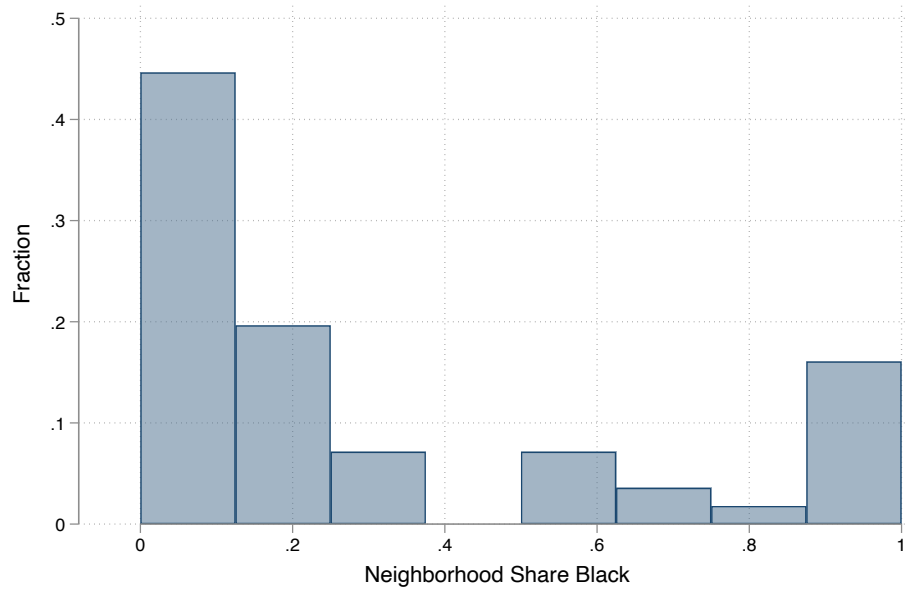


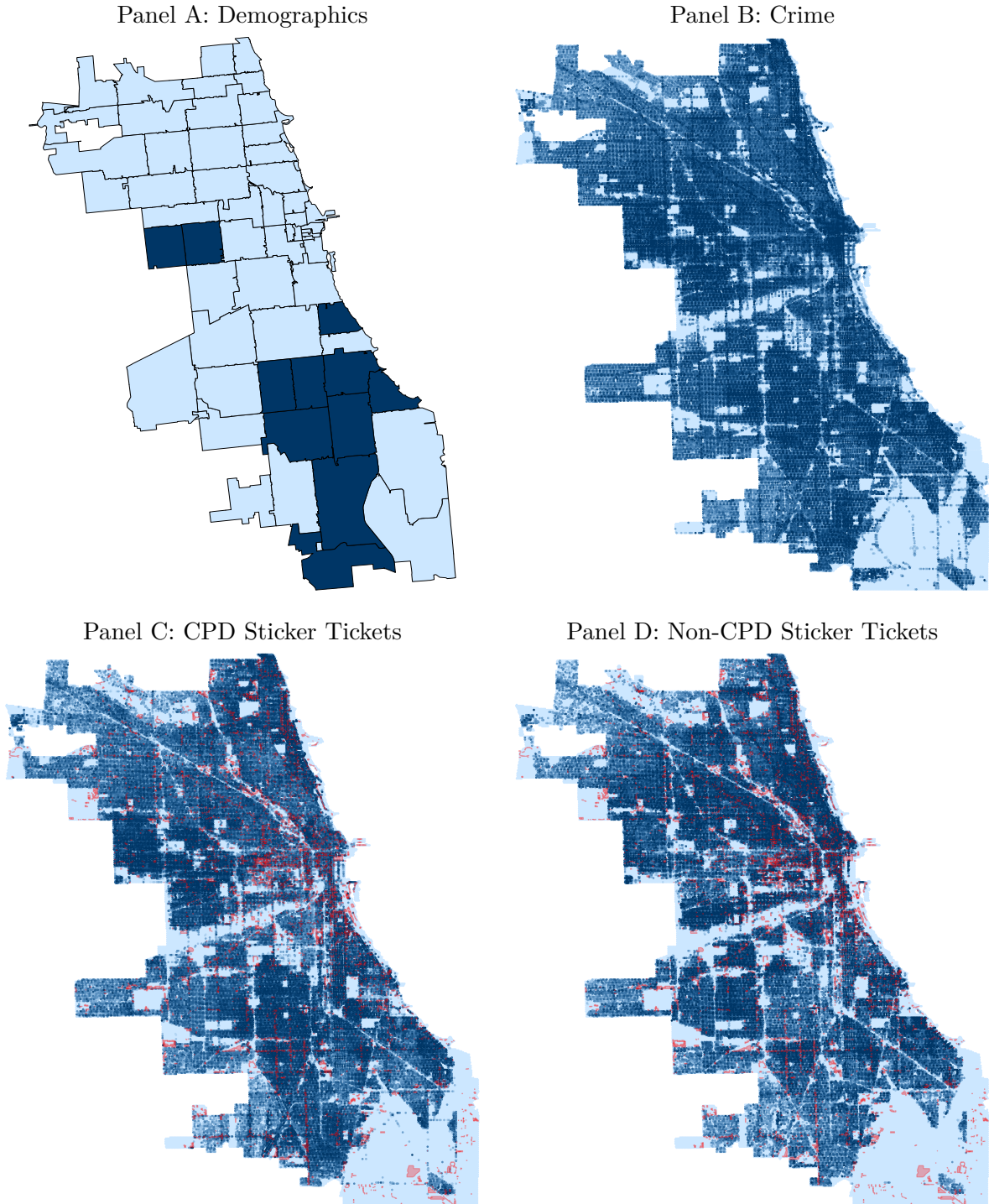
Appendix A: Additional Results

Appendix Figure A1: Distribution of Share Black Population Across Neighborhoods



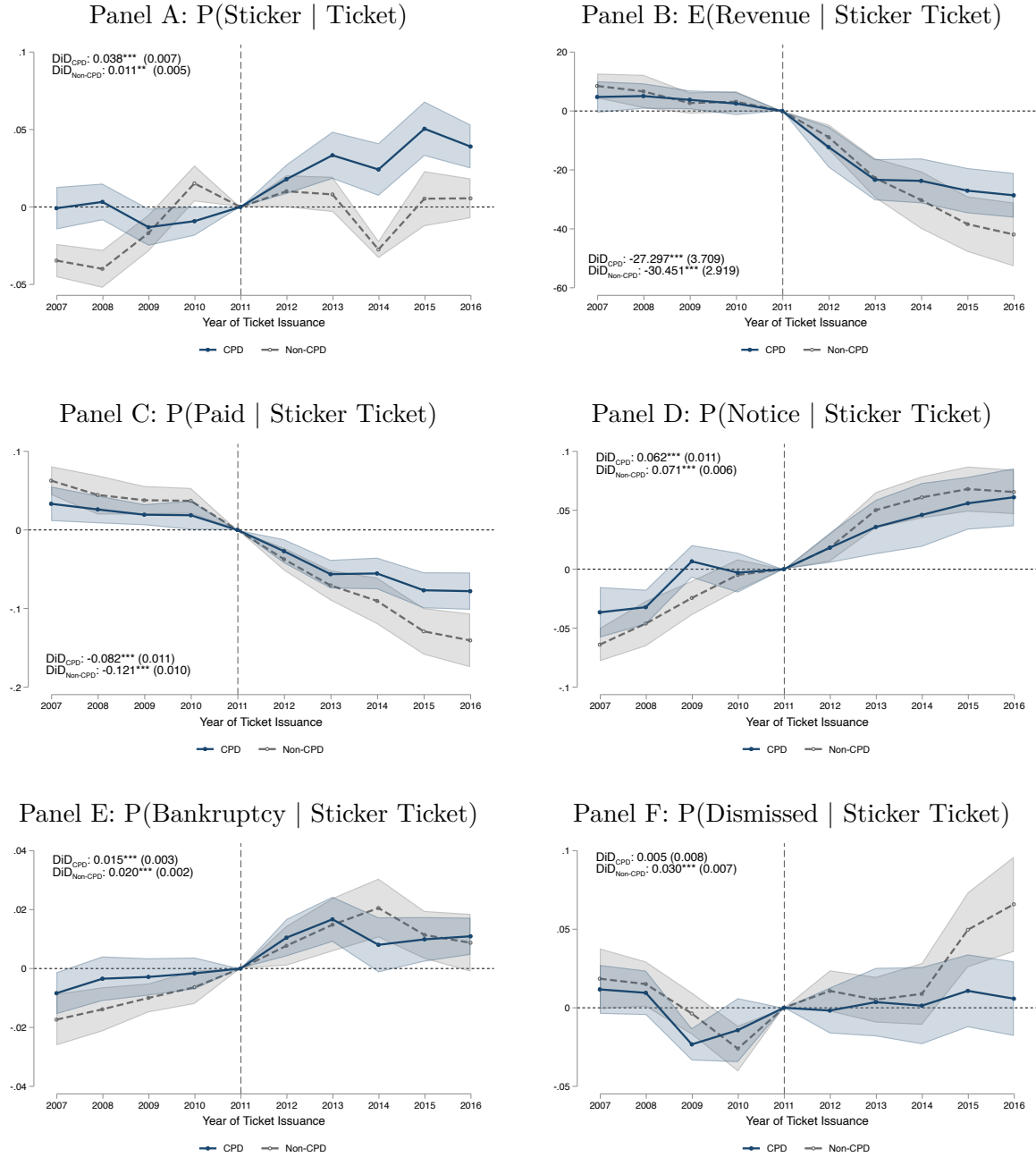
Notes: This figure reports the distribution of share Black across the neighborhoods (zip codes) in our sample. Share Black is measured using the 2007-2011 American Community Survey.

Appendix Figure A2: Maps of Sticker Tickets, Parking Amenities, Race, and Crime



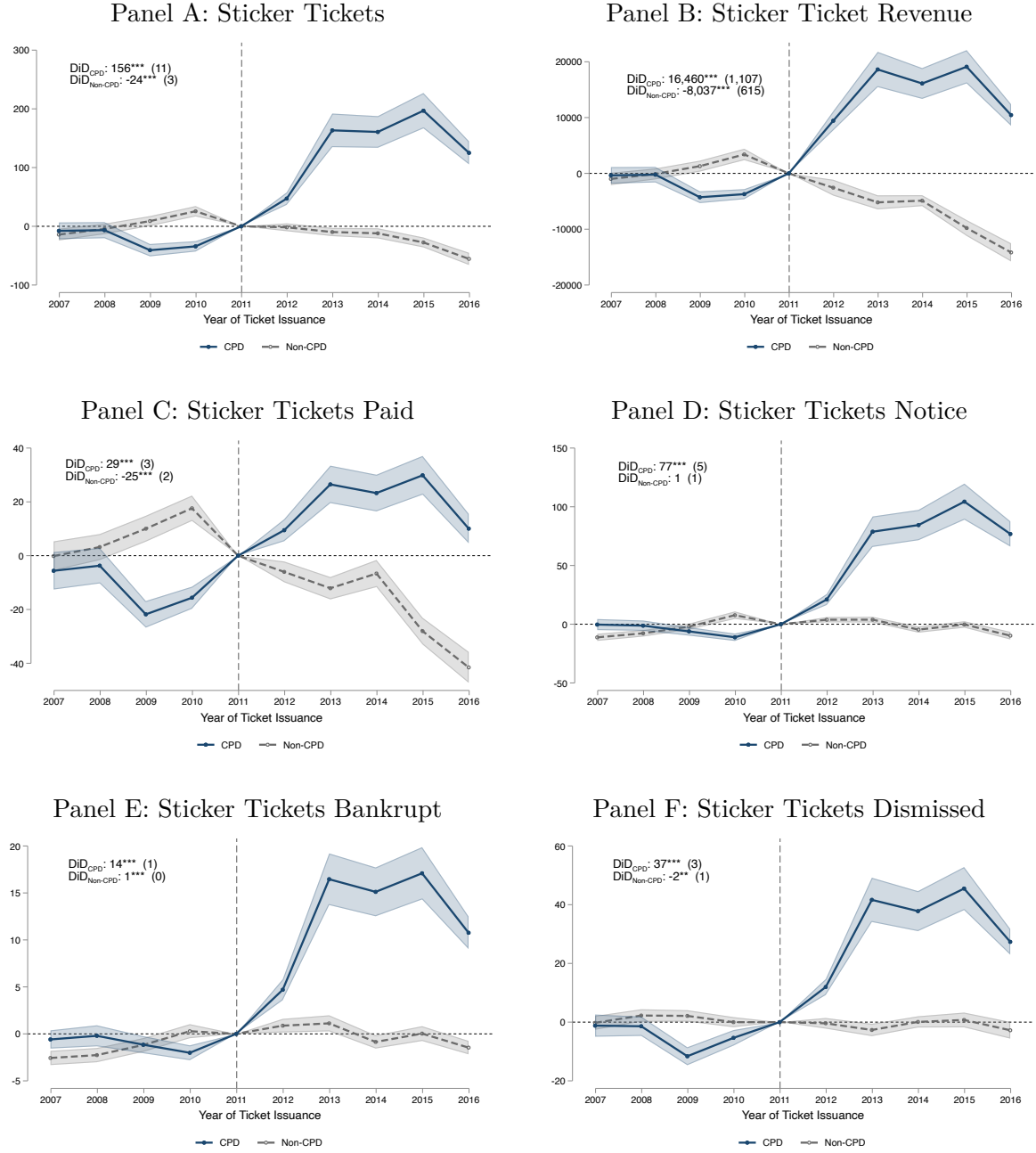
Notes: This figure reports the geographic distributions of race, crime, CPD sticker ticketing, and non-CPD sticker ticketing. Panel A shows zip codes with greater than 75% of the population reporting their race as Black (from 5 year ACS measures, 2007-2011 and 2012-2016 generate the same figure) in dark blue. Panels B-D discretize Chicago into approximately 100,000 hexagons. Density is estimated with a quartic kernel over a bandwidth of 0.0005, with each color representing a quantile. Red polygons are parking amenities defined by Open Street Map. Panel B shows the distribution of reported criminal offenses with darker Blue areas reporting higher crime. Panels C and D show sticker ticket issuance by CPD and non-CPD agents.

Appendix Figure A3: Event Study Estimates of Sticker Tickets and Outcomes at the Ticket-Level by Issuing Agency



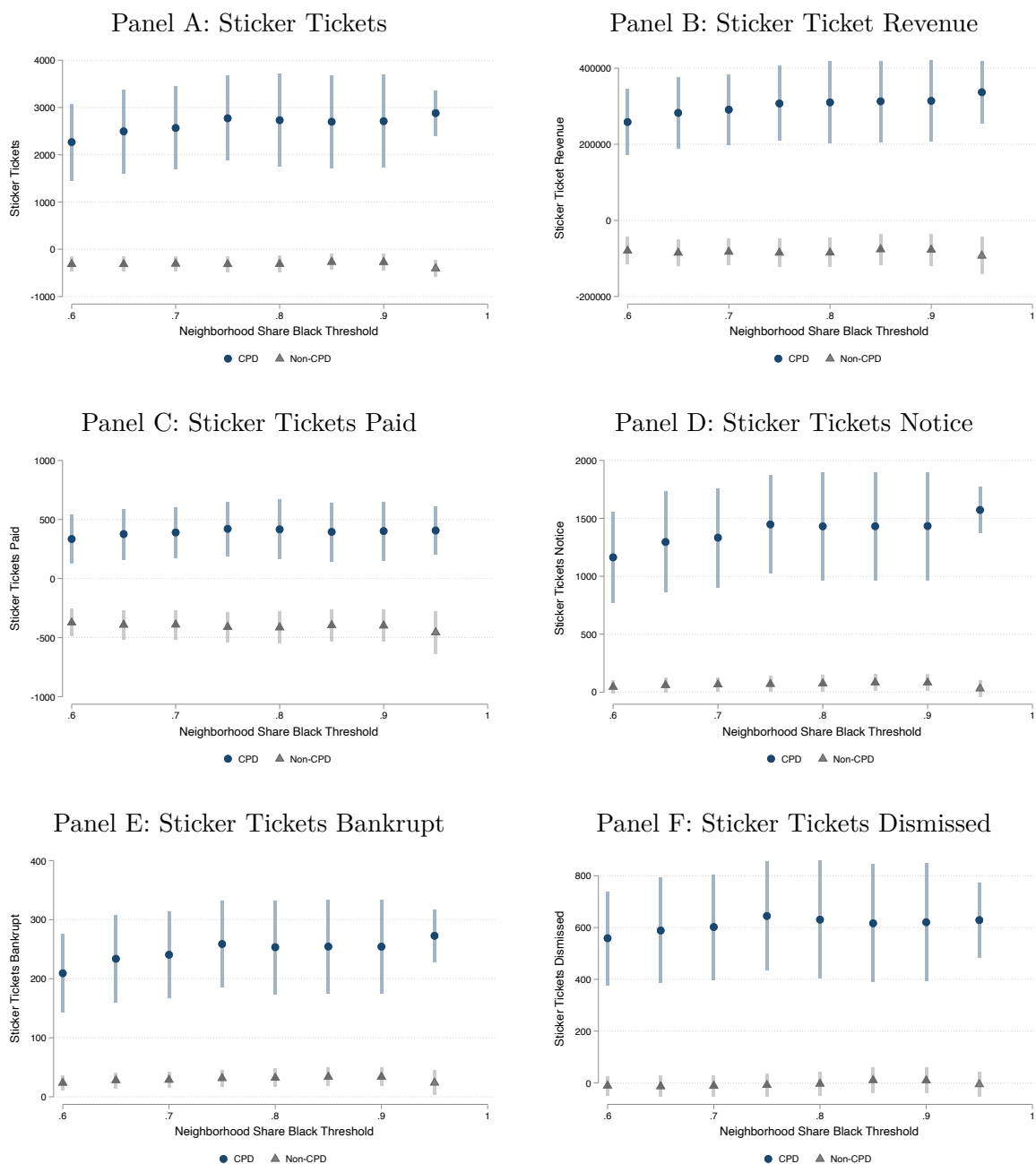
Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the ticket level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A4: Event Study Estimates of Sticker Tickets and Outcomes at the Census Tract-Level by Issuing Agency



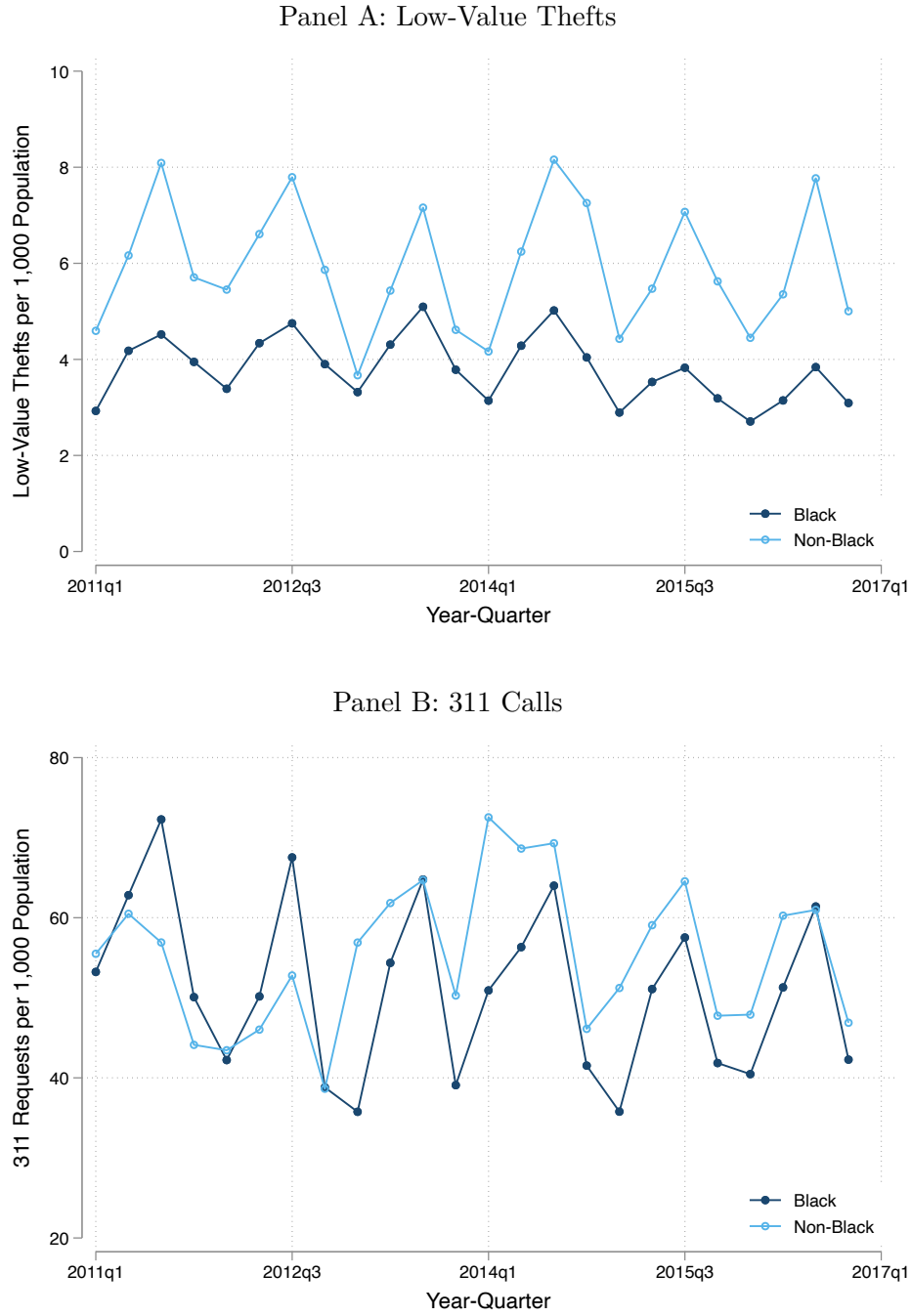
Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the neighborhood (Census tract) level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A5: Difference-in-Differences Estimates of Sticker Tickets and Outcomes by Issuing Agency, Excluding Neighborhoods in Middle of Black Share Distribution



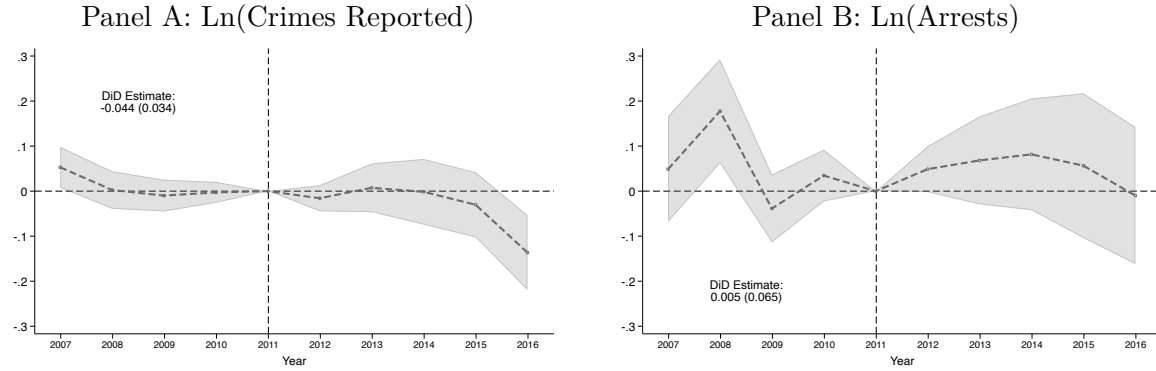
Notes: This figure reports robustness checks of our difference-in-differences (DiD) estimates for different samples. For each point, we re-define treated neighborhoods as neighborhoods with at least the x-axis share Black population (k). We then keep these neighborhoods and compare them to neighborhoods with a Black population share of less than $1 - k$, dropping neighborhoods that fall between these two cutoffs. Navy circles represent DiD estimates for CPD and gray triangles represent DiD estimates for non-CPD. Vertical shaded bars indicate 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A6: Time Series of Low-Value Thefts and 311 Calls by Neighborhood Type



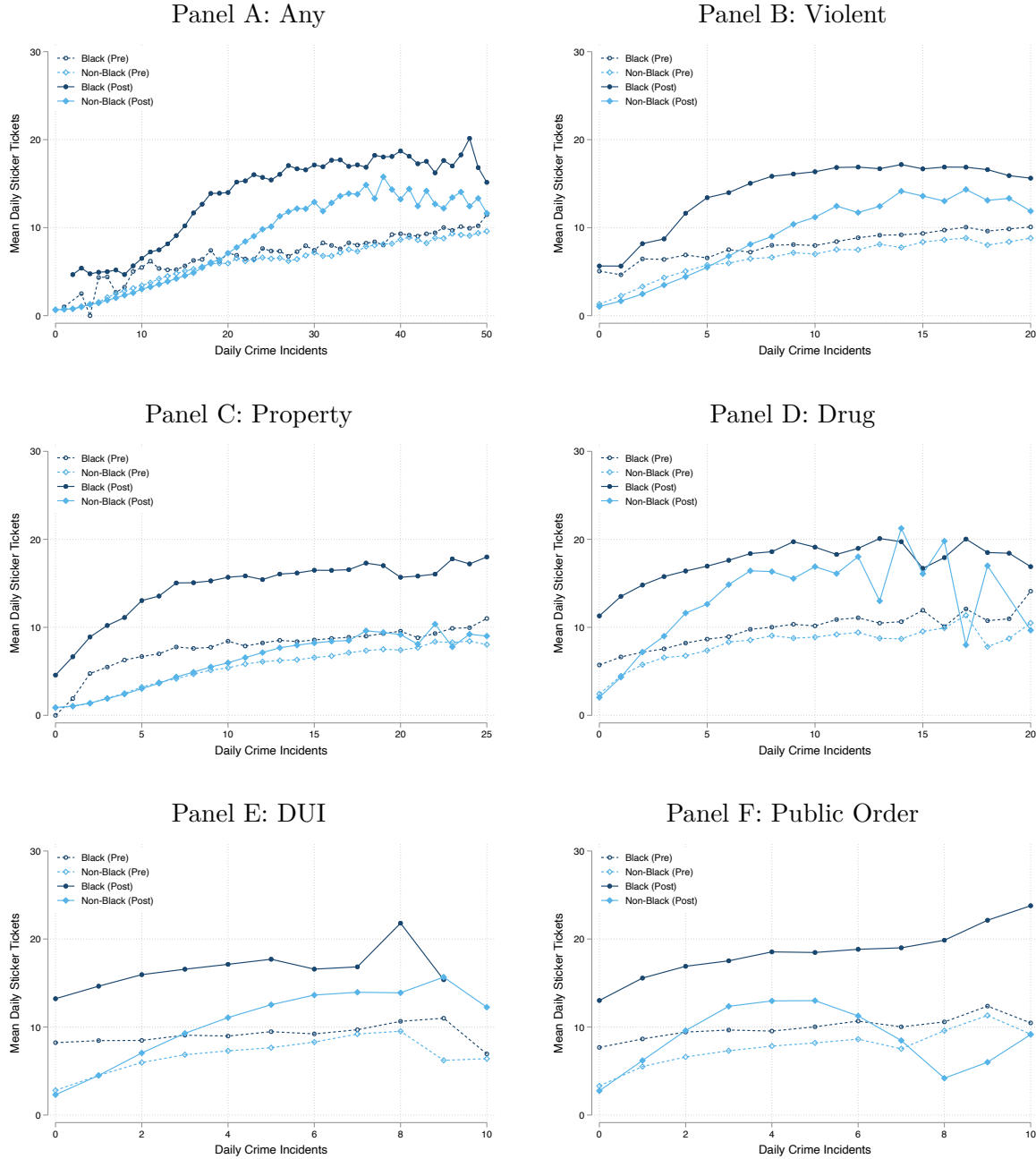
Notes: This figure reports a quarterly time series of low-value thefts (under \$500) and 311 service calls per 1,000 population. Data in Panel A come from the Chicago Police Department, data in Panel B come from available historical 311 calls from the City of Chicago. Our set of 311 calls includes: abandoned vehicles, abandoned buildings, alley lights out, garbage carts, graffiti removal, pot hole reports, rodent baiting, sanitation code complaints, street lights (all or one out), tree debris/trims, and tree trimming. We plot the average rate per 1,000 population separately by neighborhood. Solid navy dots represent Black neighborhoods and hollow light blue circles represent non-Black neighborhoods.

Appendix Figure A7: Event Study Estimates of Crimes Reported to Chicago Police Department and Arrests



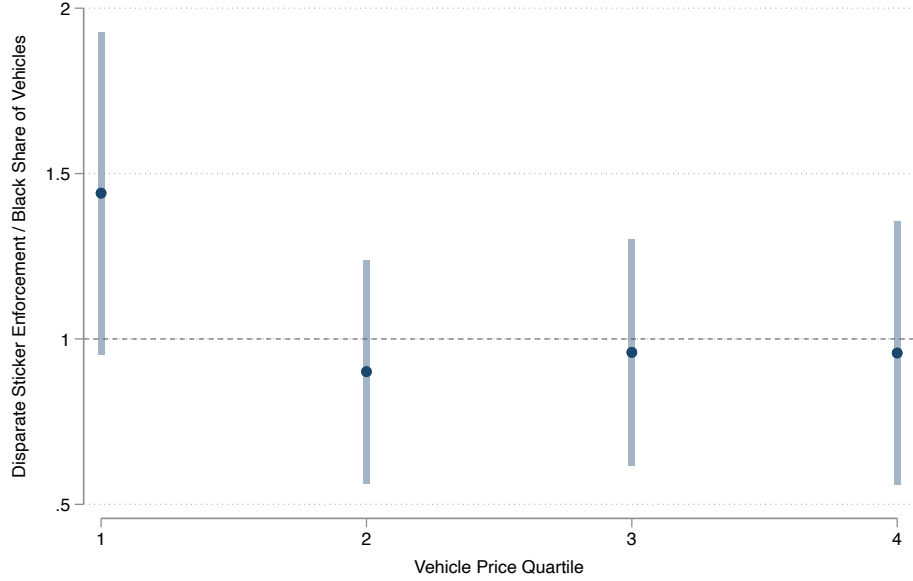
Notes: This figure reports event study estimates of crimes reported to the Chicago Police Department and arrests made for those reported offenses. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. Panel A reports results for natural log of crimes reported to the Chicago Police Department and Panel B reports results for the natural log of arrests. Corresponding difference-in-differences estimates are reported in each panel. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A8: Sticker Ticketing Over the Daily Crime Distribution



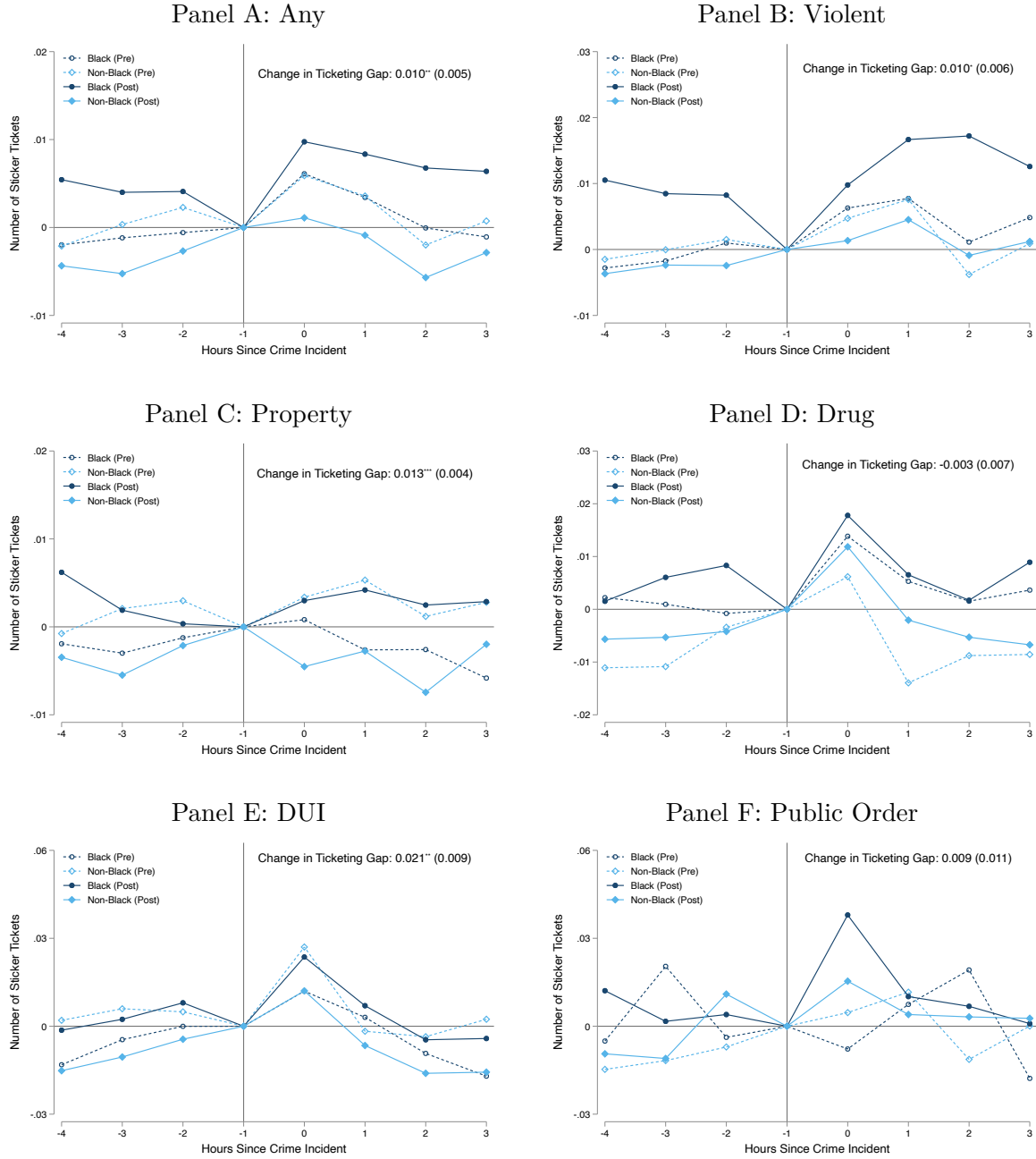
Notes: This figure reports mean daily sticker tickets over the daily crime distribution, separately for Black and non-Black neighborhoods, before and after the reform period. Each point represents the average daily number of sticker tickets written, conditional on a given daily crime level, for each neighborhood-by-reform period cell. Darker navy points depict this series for Black neighborhoods and lighter blue points depict this series for non-Black neighborhoods. Panel A reports results for all crime types and Panels B-F disaggregate total crime into crime types, classifying incidents using the machine-learning approach from Choi et al. (2023). Days with crime levels greater than the endpoint of the x-axis are grouped into the endpoint.

Appendix Figure A9: Disparate Sticker Ticket Enforcement by Vehicle Price Quartile



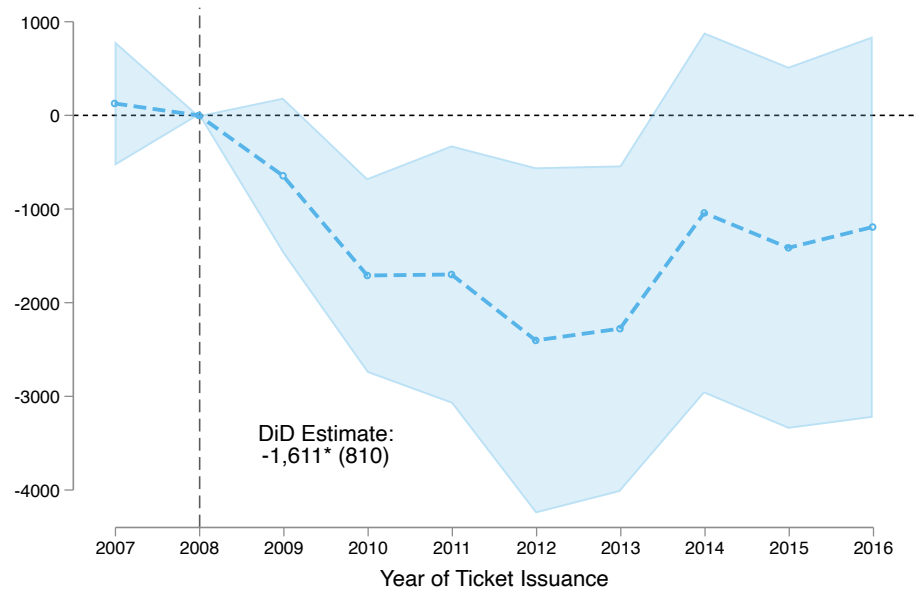
Notes: This figure reports estimate of the proportionality of the disparate sticker enforcement against vehicle price quartiles. We estimate the proportionality as the share of the disparate enforcement that accrues to vehicles in each price quartile ($\beta_q^{DiD} / \beta^{DiD}$) and then divide this quantity by the Black share of vehicles in the pre-period. Our estimate of the Black share of vehicles is based on identifying the number of unique vehicles by neighborhood from the ticketing data. Vehicle price is based on April 2025 MSRP values from Kelley Blue Book. When average MSRP value is unavailable, we assign the make the industry-wide average. Each point represents a separate calculation. Shaded bars represent 95 percent confidence intervals.

Appendix Figure A10: Event Study Estimates of Sticker Ticketing Near Criminal Incidents



Notes: This figure reports event-study estimates of sticker ticketing around the time of crimes reported to the Chicago Police Department. The outcome is the number of sticker tickets written within 300 meters of the incidents. We estimate event-study models that include the full set of interactions between event-time, neighborhood type, and reform period, and take linear combinations of these coefficients to yield the corresponding series. The model also controls for zip code fixed effects, crime type fixed effects, and date-by-hour fixed effects. For computational tractability, we use a 50% subsample of incidents. The change in ticketing gap estimates report the average change in the post-event coefficients, following our difference in differences approach in the main text. Panel A includes crimes of all types. Panels B-F estimate event-study models separately by crime types, classifying incidents using the machine learning approach from Choi et al. (2023).

Appendix Figure A11: Event Study Estimates of Parking Meter Privatization on CPD vs Non-CPD Meter Ticketing



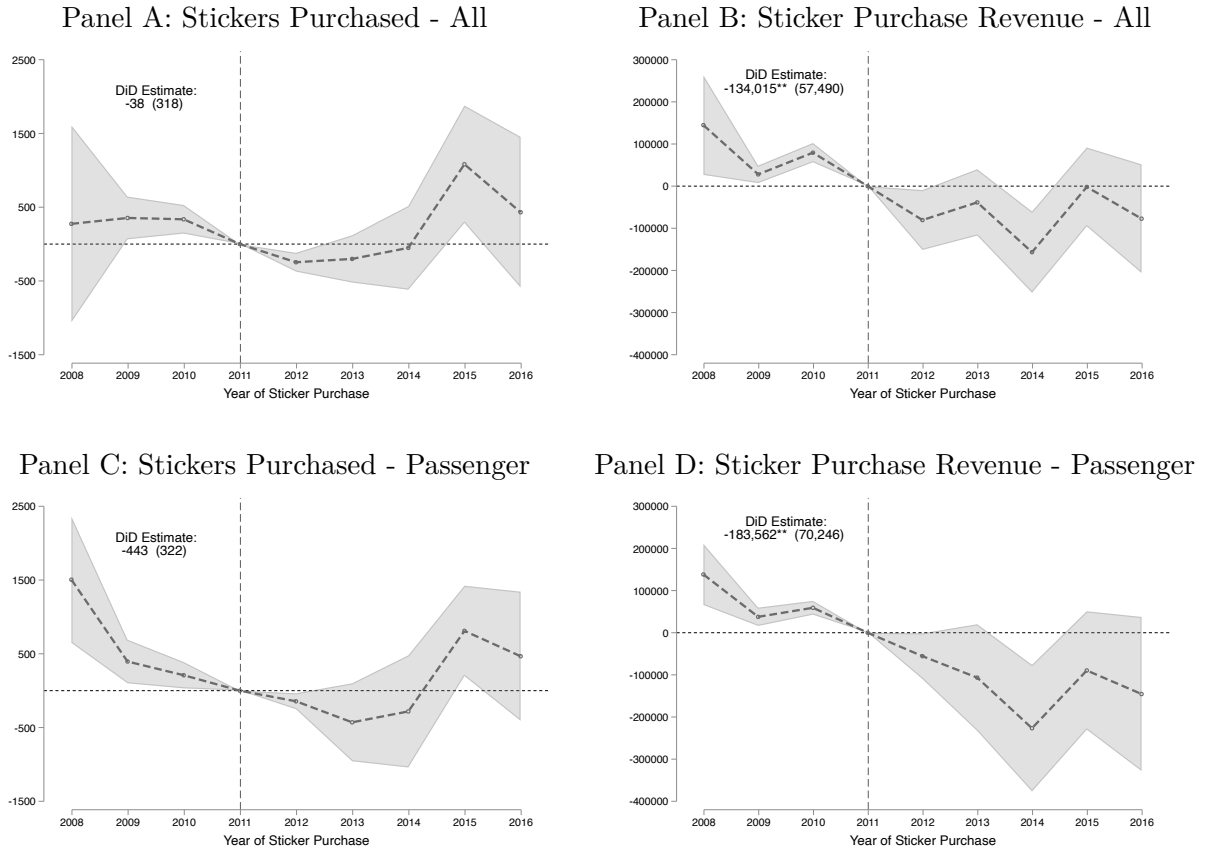
Notes: This figure reports event study estimates of parking meter tickets at the neighborhood level. Each point represents the effect of CPD parking meter ticketing activity compared to non-CPD activity, relative to 2008. The corresponding difference-in-differences estimate is displayed in the plot region. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A12: Event Study Estimates of Non-Sticker Tickets and Outcomes at the Neighborhood-Level by Issuing Agency



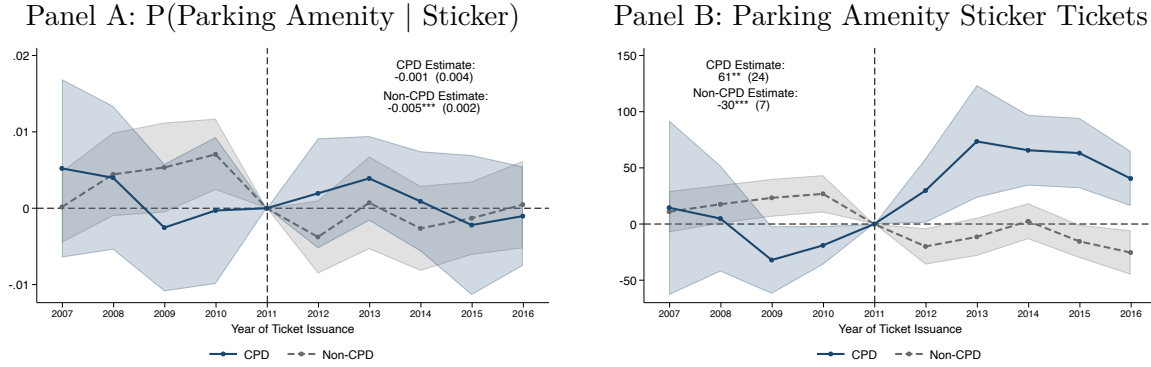
Notes: This figure reports event study estimates of non-sticker ticketing behavior and non-sticker ticket outcomes at the neighborhood level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A13: Event Study Estimates of Sticker Purchasing Behavior



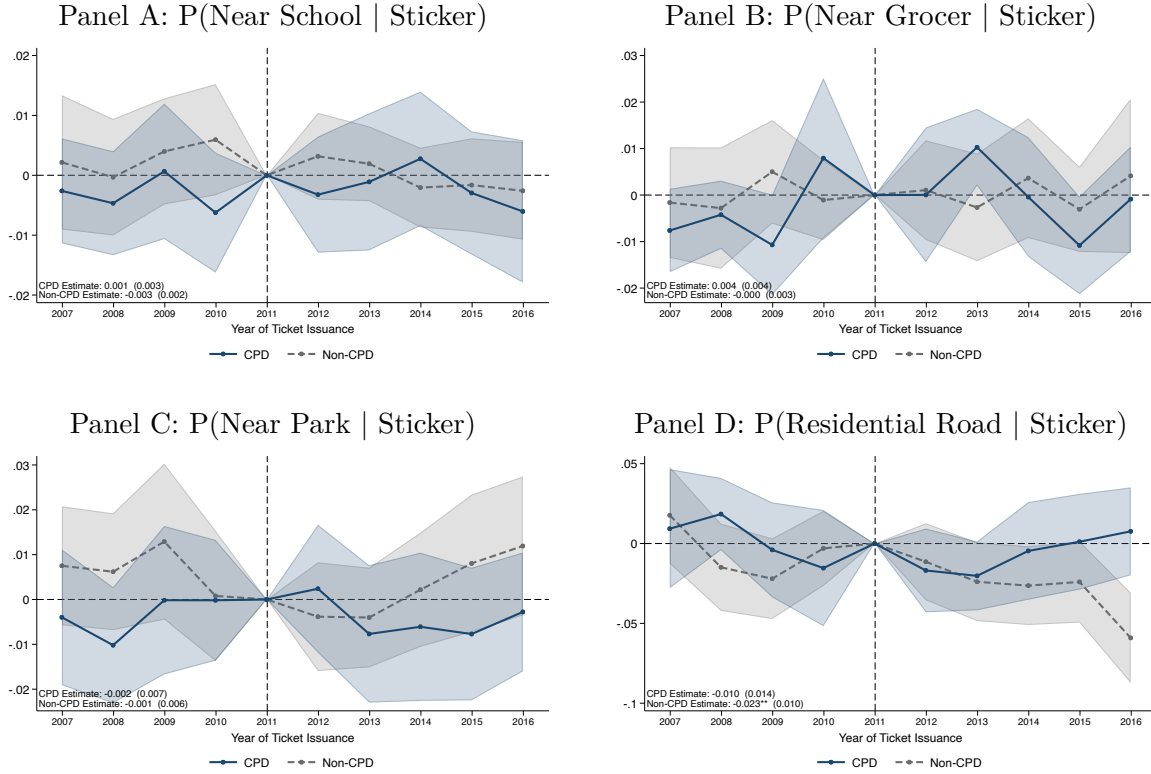
Notes: This figure reports event study estimates of sticker purchases and sticker purchase revenue at the neighborhood level. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. All includes passenger, large vehicles, and motorcycles, the latter of which we exclude from the decomposition in the lower panels. Corresponding difference-in-differences estimates are reported in each panel. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A14: Event Study Estimates of Sticker Tickets in Parking Amenities



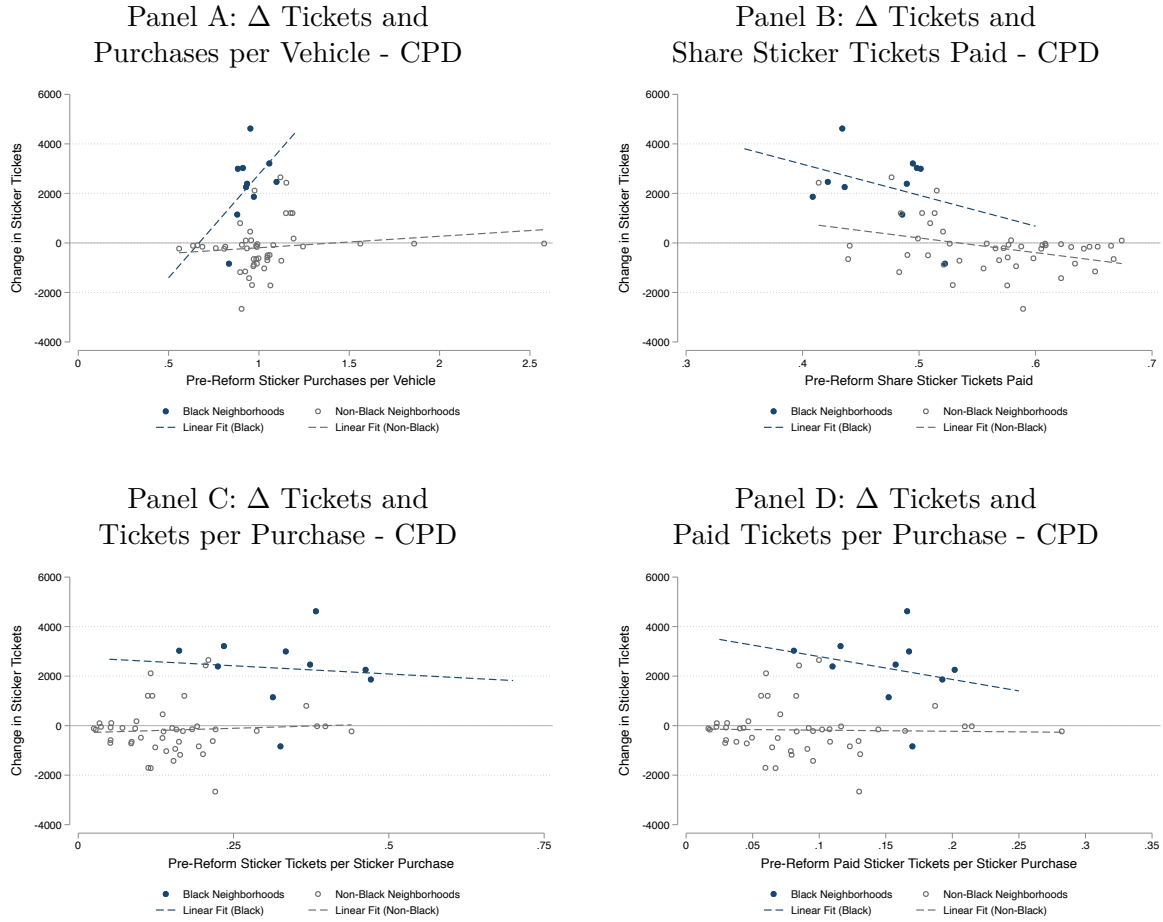
Notes: This figure reports event study estimates of sticker ticketing behavior in parking lots, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. 3.3% of CPD sticker tickets are issued in parking amenities (4.1% in non-Black zip codes, 2.2% in Black). 4.5% of not CPD sticker tickets are issued in parking amenities (5.3% in non-Black zip codes, 1.6% in Black). A ticket is deemed “in a parking amenity” if it occurs within 0.00013 lat/lon (about 15 meters) from a parking amenity or within a parking amenity as defined by Open Street Mapping. Results are robust to alternative buffer regions, even at a buffer of 0.0002, fewer than 9% of sticker tickets are in parking amenities. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A15: Event Study Estimates of Sticker Tickets by Location



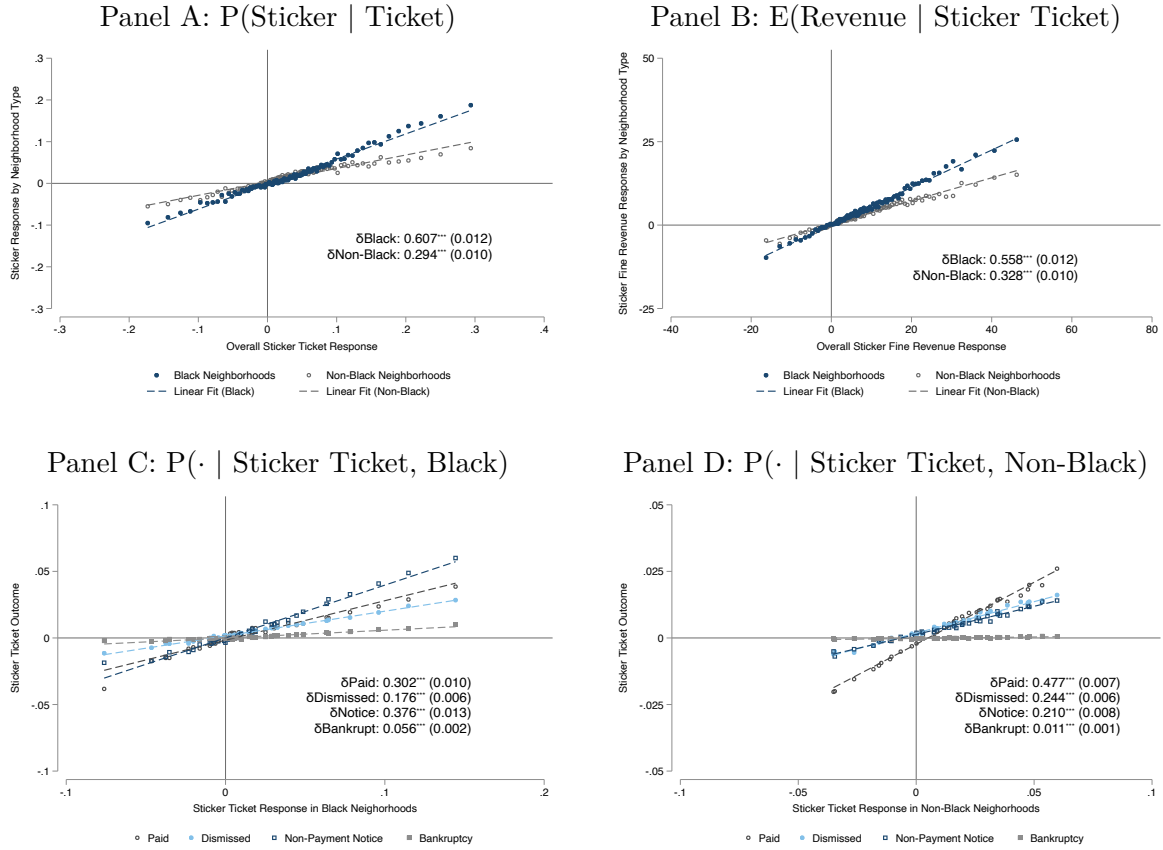
Notes: This figure reports event study estimates of sticker ticketing behavior in parking lots, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. A ticket is deemed near a school, grocer, or park if the ticket was issued within 100 meters of that amenity as defined by Open Street Mapping. Amenities are coded from OpenStreetMap tags as follows: schools (amenity=school), parks (leisure=park), and grocers (shop in supermarket, convenience, grocery, greengrocer). Distances are computed in meters after projecting coordinates to EPSG; school and grocer polygons are converted to centroids, while parks are kept as polygons (so distance is to the nearest park boundary). Roads are classified from the OSM drivable network. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A16: Additional Relationships Between Change in Sticker Ticketing and Neighborhood Characteristics



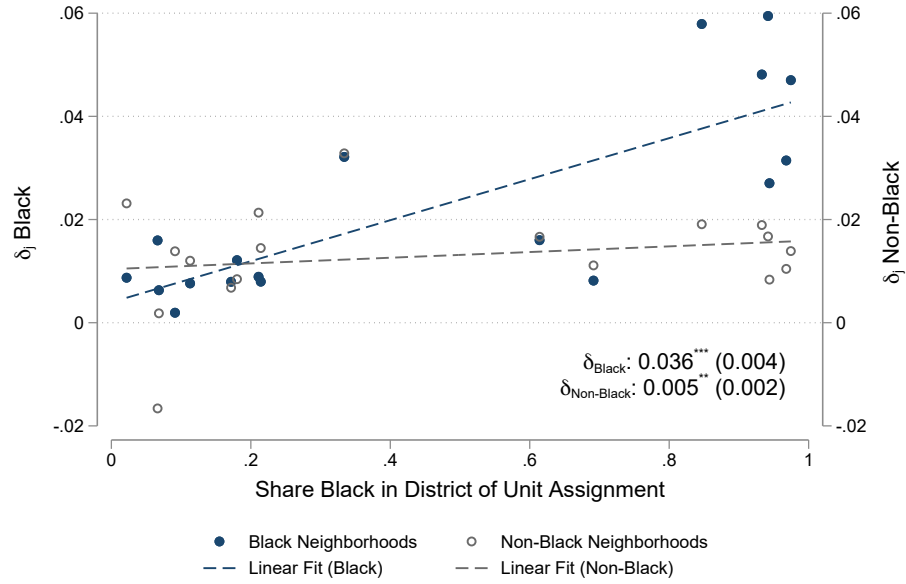
Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates across different outcomes, along with neighborhood-level estimates with pre-reform characteristics. Each panel plots the change in sticker tickets issued by CPD against pre-reform neighborhood characteristics, calculated using data from 2008-2011. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

Appendix Figure A17: Estimating and Decomposing Officer-Specific Responses to Sticker Fine Increase - Empirical Bayes-Adjusted



Notes: This figure plots Empirical Bayes-adjusted estimates of δ_j for different outcomes against each other. In Panels A and B we plot estimates of neighborhood race-specific δ_j responses against the overall race-agnostic δ_j response on the x-axis. In Panels C and D, we plot the race-specific sticker ticket outcomes against the race-specific sticker ticket responses, separately by race. There are 100 bins per outcome in the top two panels and 40 bins per outcome in the lower two panels. For exposition, we drop the first and last bin for each outcome. Reported coefficients estimated on the underlying officer-level estimates. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A18: Officer-Specific Policy Responses and Assignment Demographics



Notes: This figure reports the correlation between officer-specific δ_j sticker ticket responses by neighborhood demographic group and the demographic composition of the modal unit assignment. Each point represents a separate unit assignment and plots the within-bin mean against share Black. Blue dots denote $\delta_j(Sticker, Black)$ responses (left axis) and gray circles represent $\delta_j(Sticker, Non - Black)$ responses (right axis). Dashed lines denote linear fits. Reported coefficients and standard errors are estimated on the underlying data. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A1: Annual Volume and Revenue from Top Ten Revenue Generating Tickets in Chicago by Neighborhood
Demographic Composition

	2007-2011						2012-2016					
	Tickets	Revenue	Avg.	Rev.	Rev.	Rev./	Tickets	Revenue	Avg.	Rev.	Rev.	Rev./
	(000s)	(\$000s)	(3)	Share	Rank	E[Rev.]	(000s)	(\$000s)	Fine	Share	Rank	E[Rev.]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel A: Black Neighborhoods</i>												
No City Sticker	48	5,034	120	0.24	1	0.87	71	8,593	200	0.33	1	0.60
Expired Plates	79	3,631	50	0.18	2	0.91	88	4,176	60	0.16	2	0.79
Street Cleaning	34	1,816	50	0.09	3	1.07	33	1,901	60	0.07	3	0.96
Rear And Front Plate Required	32	1,345	50	0.06	4	0.85	28	991	60	0.04	8	0.59
Park In Transit Stand	13	1,239	100	0.06	5	0.98	12	1,123	100	0.04	5	0.95
Residential Permit Parking	18	1,076	60	0.05	6	0.98	15	1,047	75	0.04	6	0.92
Parking Prohibited Anytime	15	913	60	0.04	7	0.99	20	1,419	75	0.05	4	0.96
Within 15' Of Fire Hydrant	8	800	100	0.04	8	0.98	9	1,034	150	0.04	7	0.76
Rush Hour Parking	14	796	60	0.04	9	0.97	9	662	100	0.02	10	0.77
Expired Meter	11	566	50	0.03	10	1.02	8	426	50	0.02	14	1.02
<i>Panel B: Non-Black Neighborhoods</i>												
Expired Meter	393	18,966	50	0.16	1	0.97	274	14,839	50	0.12	3	1.08
No City Sticker	129	13,580	120	0.12	2	0.88	134	20,982	200	0.17	1	0.78
Expired Plates	288	13,520	50	0.12	3	0.94	274	15,386	60	0.13	2	0.94
Street Cleaning	248	13,004	50	0.11	4	1.05	238	13,865	60	0.12	4	0.97
Residential Permit Parking	204	11,782	60	0.10	5	0.96	164	11,986	75	0.10	5	0.97
Parking Prohibited Anytime	129	7,133	60	0.06	6	0.92	109	7,851	75	0.07	6	0.96
Rush Hour Parking	108	6,327	60	0.05	7	0.98	59	4,871	100	0.04	8	0.83
Rear And Front Plate Required	122	5,437	50	0.05	8	0.89	89	3,631	60	0.03	10	0.68
Within 15' Of Fire Hydrant	38	3,749	100	0.03	9	0.99	34	4,085	150	0.03	9	0.81
Park In Transit Stand	39	3,667	100	0.03	10	0.94	26	2,465	100	0.02	11	0.95

Notes: This table reports average annual characteristics of the top ten revenue generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016, separately by neighborhood demographic composition. Panel A reports statistics for Black neighborhoods, define as neighborhoods whose population is at least seventy-five percent Black, and Panel B reports statistics for Non-Black neighborhoods. The type of ticket is listed in each row. Columns 1 and 7 report annual ticket volume, Columns 2 and 8 report annual ticket revenue, Columns 3 and 9 report the modal fine amount, Columns 4 and 10 report the revenue share of the listed ticket, along with the revenue share rank in Columns 5 and 11. Columns 6 and 12 report the ratio of revenue received and expected revenue, calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time, and is greater than one if collected revenue plus applicable late fees exceeds this expectation.

Appendix Table A2: Robustness Checks of Neighborhood-Level Regressions

<i>Panel A: CPD</i>	(1)	(2)	(3)	(4)	(5)
Tickets	2,490*** (461)	2,490*** (464)	2,308*** (414)	2,166*** (385)	2,347*** (508)
Revenue	271,341*** (50,779)	271,341*** (51,102)	256,656*** (47,257)	249,193*** (42,036)	260,740*** (56,236)
Paid	363*** (113)	363*** (114)	283*** (98)	355*** (100)	336*** (122)
Notice	1,316*** (220)	1,316*** (222)	1,279*** (198)	1,088*** (189)	1,249*** (244)
Bankruptcy	236*** (38)	236*** (38)	222*** (33)	195*** (33)	222*** (42)
Dismissed	574*** (107)	574*** (108)	522*** (102)	528*** (87)	539*** (117)
<i>Panel B: Non-CPD</i>					
Tickets	-294*** (82)	-294*** (83)	-392*** (133)	-323*** (78)	-314*** (85)
Revenue	-77,131*** (17,194)	-77,131*** (17,303)	-83,071*** (17,613)	-80,922*** (17,588)	-80,994*** (17,189)
Paid	-374*** (64)	-374*** (65)	-358*** (95)	-371*** (58)	-383*** (68)
Notice	58* (34)	58* (34)	-23 (32)	40 (26)	53 (37)
Bankruptcy	29*** (7)	29*** (7)	3 (11)	23*** (6)	28*** (8)
Dismissed	-9 (20)	-9 (20)	-16 (15)	-14 (18)	-12 (21)
Zip Fixed Effects	No	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	C-S	OLS	OLS
Black Majority Cutoff	0.75	0.75	0.75	0.50	0.90

Notes: This table reports robustness checks of our neighborhood difference-in-differences (DiD) estimates. Panel A reports results for CPD-written tickets and Panel B reports results for non-CPD-written tickets. The outcome is listed in each row. Each cell corresponds to a different point estimate. Column 1 estimates the simple DiD without any additional fixed effects, Column 2 reproduces our main text estimate, Column 3 uses the estimator from Callaway and Sant'Anna (2021), Column 4 changes the cutoff to define Black neighborhoods to 50%, and Column 5 changes the same threshold to 90%. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A3: Decomposing Differential Outcomes by Owner Zip Code Demographics

	Tickets	Revenue	Paid	Notice	Bankruptcy	Dismissed
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: CPD</i>						
Main:	2,490*** (464)	271,341*** (51,102)	363*** (114)	1,316*** (222)	236*** (38)	574*** (108)
Black Zip Owner:	1,464*** (278)	196,497*** (32,861)	53 (66)	968*** (151)	184*** (25)	257*** (43)
Non-Black Zip Owner:	772*** (179)	65,194*** (21,376)	269*** (57)	323*** (85)	48*** (15)	132*** (36)
<i>Panel B: Non-CPD</i>						
Main:	-294*** (83)	-77,131*** (17,303)	-374*** (65)	58* (34)	29*** (7)	-9 (20)
Black Zip Owner:	-83* (46)	25,558*** (7,683)	-183*** (41)	75*** (24)	24*** (5)	1 (7)
Non-Black Zip Owner:	-212*** (46)	-96,605*** (12,785)	-175*** (31)	-15 (13)	5* (3)	-27*** (10)

Notes: This table decomposes our main difference-in-differences estimates into outcomes experienced by owners in majority (>75 percent) Black neighborhoods and those in non-Black majority neighborhoods. We interact each outcome in the column title with indicators for $Black_i$ and $(1 - Black_i)$. The “Main” row reproduces our main text estimate and the corresponding “Black” and “Non-Black” rows decompose the Main outcome following the previous description. Due to missing owner information for some tickets, the decomposition will not exactly add to the full sample estimate. Panel A reports results for CPD-written tickets, and Panel B reports results for non-CPD-written tickets. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A4: Decomposing Differential Outcomes by Non-Race Neighborhood Characteristics Within Black Neighborhoods

	Crime Rate		Income		Sticker Ticket Rate		Sticker Ticket Payment Rate	
	High	Low	High	Low	High	Low	High	Low
<i>Panel A: CPD</i>								
Sticker	3,017*** (463)	1,964*** (699)	3,055*** (537)	1,925*** (630)	2,122*** (686)	2,858*** (540)	2,334*** (710)	2,647*** (552)
<i>p</i> -value	0.200		0.164		0.389		0.721	
<i>Panel B: Non-CPD</i>								
Sticker	-328*** (75)	-261** (126)	-338*** (106)	-251** (99)	-309*** (97)	-280** (112)	-284** (117)	-305*** (91)
<i>p</i> -value	0.600		0.489		0.822		0.868	

Notes: This table reports difference-in-differences results which decompose the differential response in Black neighborhoods along other non-race characteristics, estimated at the neighborhood-level. Each non-race characteristic is defined in the pre-reform period and splits the subsample of Black neighborhoods into above- and below-median groups based on the statistic listed in the column title. Sticker ticket rate is the fraction of neighborhood tickets which are sticker tickets. Sticker ticket payment rate is the fraction of neighborhood sticker tickets which are paid. Panel A reports results for CPD-written tickets and Panel B reports results for non-CPD-written tickets. Listed p-values test for differences between coefficient estimates in Black neighborhoods. All regressions include zip code and year fixed effects. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A5: Decomposing Relative Contributions of Differential Compliance and Enforcement in Sticker Ticketing Gap

	Black Neighborhoods	Non-Black Neighborhoods	Gap
	(1)	(2)	(1)-(2)
Share Vehicles Unregistered	0.196	0.200	-0.004
Δ Enforcement Share	0.689	-0.047	0.737
Δ Compliance Share	0.123	0.121	0.002
Total Vehicles	17,320	21,893	-4,574
Unregistered Vehicles	3,356	3,725	-369

Notes: This table calculates statistics describing the size of the unregistered vehicle stock in the pre-reform period, separately by neighborhood type. The first row calculates the share of vehicles without a valid sticker as of the end of the grace period. The number of unregistered vehicles is estimated using the 2007-2011 American Community Survey. We drop a small handful of neighborhoods with negative estimated non-compliance rates due to measurement error in number of vehicles. The second and third rows estimate the share of unregistered vehicles who were affected by the reform, dividing one-way differences in CPD enforcement and consumer purchases by the number of unregistered vehicles. Column 1 reports these estimates for Black neighborhoods, Column 2 for non-Black neighborhoods, and Column 3 reports the difference between Columns 1 and 2.

Appendix Table A6: Correlating Officer Policy Responses with Observable Characteristics
- All Officers

	δ_j Response			
	(1)	(2)	(3)	(4)
<i>Panel A: δ_j (Sticker, Black)</i>				
Male	0.001 (0.003)			0.001 (0.003)
Age	-0.000** (0.000)			-0.000 (0.000)
Hispanic		-0.001 (0.003)		-0.001 (0.003)
Asian or Native American		-0.002 (0.006)		-0.003 (0.006)
Black		-0.008** (0.004)		-0.007* (0.004)
Years Experience			-0.000** (0.000)	-0.000 (0.000)
Complaints per Year			0.000 (0.002)	-0.000 (0.002)
Tickets Issued per Year (00s)			0.001* (0.001)	0.001 (0.001)
<i>Panel B: δ_j (Sticker, Non-Black)</i>				
Male	-0.000 (0.002)			-0.001 (0.002)
Age	-0.000 (0.000)			-0.000 (0.000)
Hispanic		0.001 (0.002)		0.001 (0.002)
Asian or Native American		0.005 (0.005)		0.005 (0.005)
Black		0.000 (0.002)		0.001 (0.002)
Years Experience			0.000 (0.000)	0.000 (0.000)
Complaints per Year			0.001 (0.001)	0.001 (0.001)
Tickets Issued per Year (00s)			0.001 (0.001)	0.001 (0.001)
Observations	6,153	6,153	6,153	6,153
Unit Fixed Effects	Yes	Yes	Yes	Yes

Notes: This table reports regressions of officer-specific δ_j responses against officer-level observables. The sample includes all police officers. The dependent variable in Panel A is the officer-specific δ_j for sticker tickets in Black neighborhoods and the dependent variable in Panel B is the corresponding δ_j for sticker tickets in Non-Black neighborhoods. Experience, complaints and tickets issued per year are all measured prior to the policy change. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix B: Model of Officer Behavior

In this section, we present a simple model of CPD and non-CPD officer behavior to provide a simple microfoundation for understanding the differences between officers across departments and illustrating how the reform impacts various channels which influence officer behavior.

Non-CPD Behavior: We operationalize non-CPD agencies as maximizing ticket volume:

$$\max_{s_n} s_n v_n - \underbrace{c(s_n)}_{\text{Search cost}}$$

where s_n is a search in neighborhood n , v_n is the violation rate in neighborhood n , and $c(s_n) = c_n s_n^2/2$ is the neighborhood-specific convex search cost. Standard first order conditions equate the violation rate with the marginal cost of search. Notably, this solution is independent of the expected value of the ticket, including both the fine amount and payment probability.

CPD Behavior: In contrast, CPD maximizes revenue and other objectives:

$$\max_{s_n} s_n \underbrace{(f * p_n * v_n)}_{\text{E[Revenue]}} - \underbrace{c(s_n)}_{\text{Search cost}}$$

In this setup $\text{E[Revenue]} = (f * p_n * v_n)$, where f is the fine, p_n is the probability a ticket gets paid and $c(s_n) = c_n s_n^2/2$ is the cost of search, including the opportunity cost.

The first order conditions of this problem should equate marginal expected revenue to marginal costs. Thus, there will more tickets in neighborhood n if, all else equal there are higher violation rates, higher payment rates, or lower marginal costs. The fine increase will affect both the level and distribution of tickets. Writing out the FOCs:

$$\underbrace{f * p_n * v_n}_{\text{Marginal revenue}} = \underbrace{c_n s_n^*}_{\text{Marginal cost}}$$

We are interested in the change in search with respect to a change in fine, $\frac{\delta s_n^*}{\delta f}$, implying:

$$\underbrace{p_n * v_n}_{\text{Revenue effect}} + \underbrace{f * \frac{\delta p_n}{\delta f} * v_n}_{\text{Payment rate effect}} + \underbrace{f * p_n * \frac{\delta v_n}{\delta f}}_{\text{Deterrence effect}} = c_n \frac{\delta s_n^*}{\delta f}$$

The first term of this equation ($p_n * v_n$) can be thought of as a revenue effect, which we find as smaller in Black neighborhoods. The second term, the payment rate effect, is likely smaller in Black neighborhoods because of a larger reduction in repayment, with low violation rates across all neighborhoods. The final term, we can label a deterrence effect, which we find is relatively small. We conclude that the larger increase in tickets in Black neighborhoods is likely due to smaller marginal costs for CPD officers.